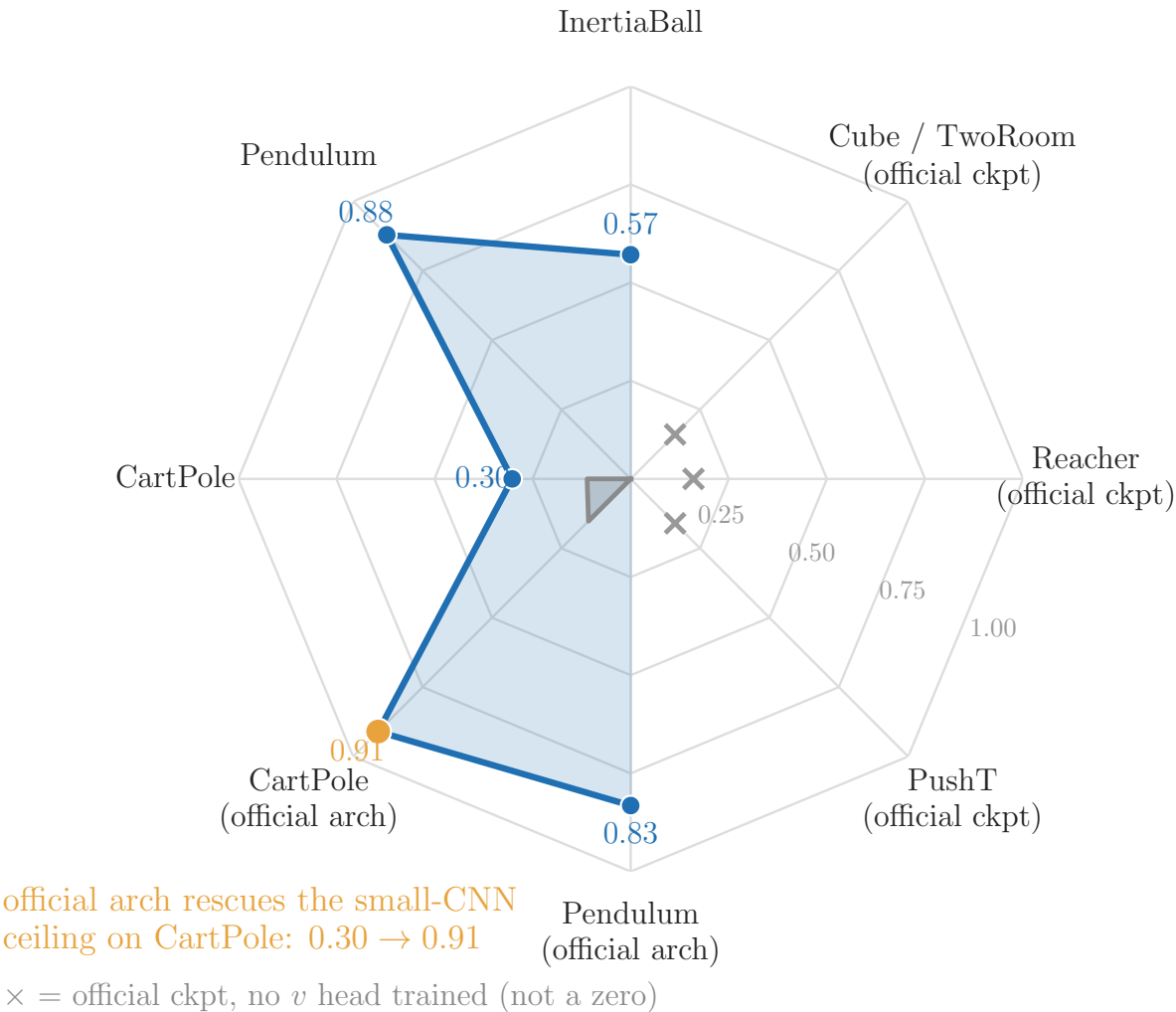


Target identifiability: a single-frame JEPA never sees  $\dot{q}$ ; a pose/motion split does.

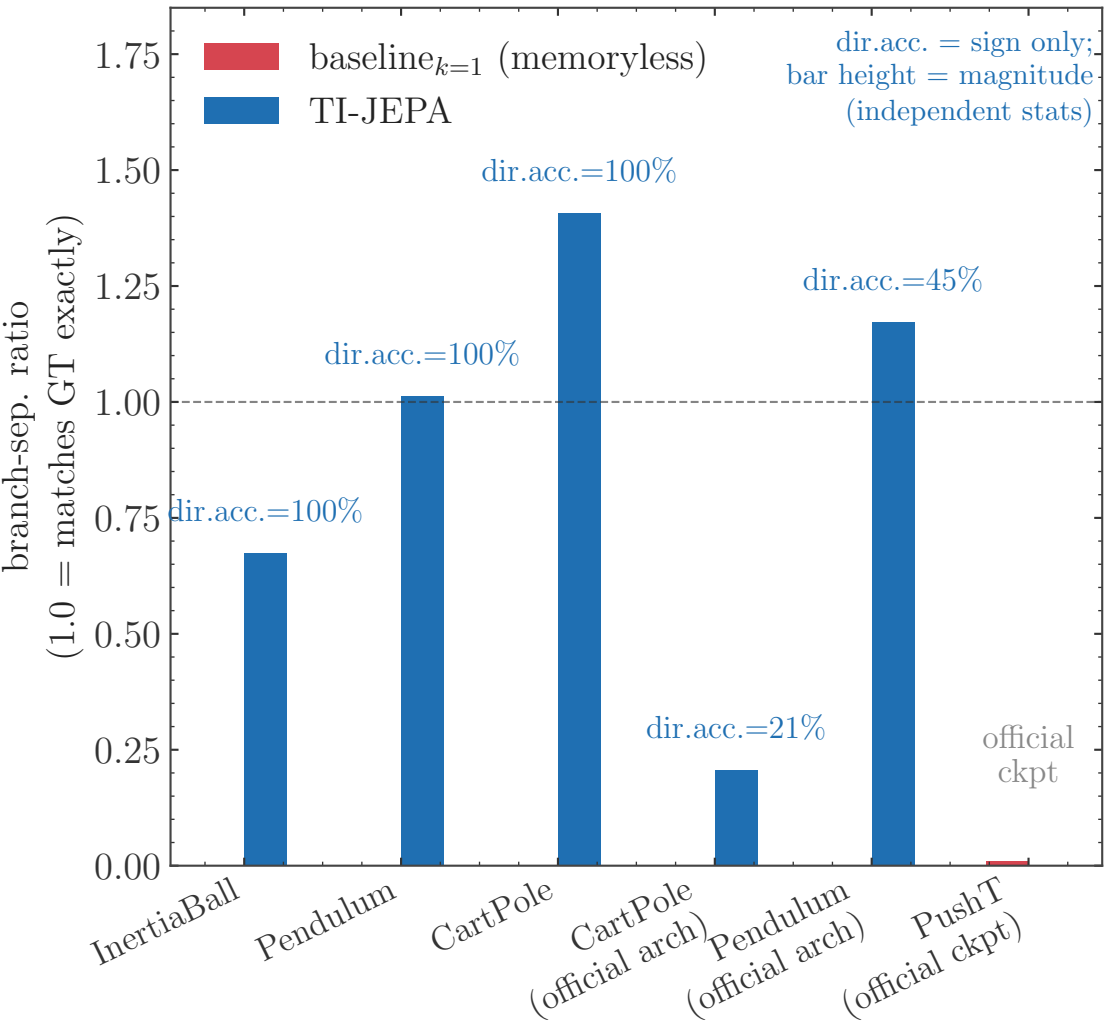
A Protocol A — probe ladder

single-frame  $\approx 0$  everywhere — explicit  $v_t$  is what was missing



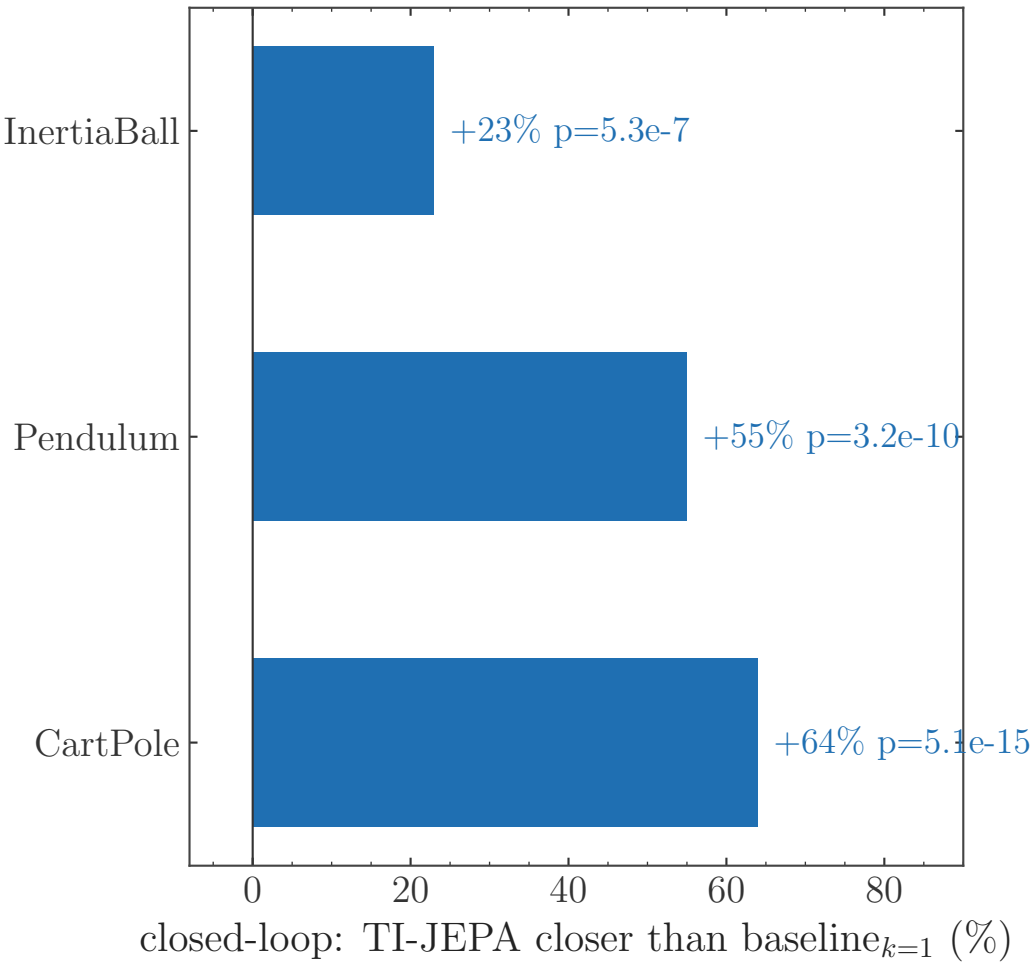
B Protocol B — kill experiment

same  $q$ , opposite  $v$ : memoryless model always collapses to 0



C Protocol C — planning (matched memory)

wins only where velocity is necessary and the model can represent it



**Custom envs:** one small CNN, one recipe. **Official arch:** from-scratch ViT-Tiny/14 + AdaLN transformer, official recipe. **Official ckpt:** released LeWM weights, zero retraining.

— single-frame  $z_t / q_t$  — TI-JEPA explicit  $v_t$